

2024 HSC Mathematics Advanced

Marking Guidelines

Section I

Question	Answer
1	A
2	C
3	B
4	B
5	C
6	B
7	D
8	B
9	D
10	C

Question 1

$$0.25^3 = 0.0015625 \Rightarrow 1.56\%$$

Therefore, option A.

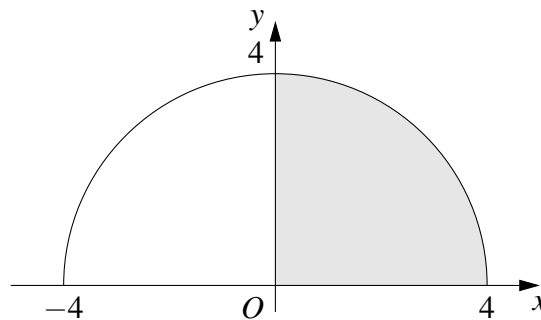
Question 2

Both options C and D appear to have a negative correlation coefficient. However, the data in option D shows a near perfect line which would have a correlation coefficient of approximately -1 .

Therefore, option C.

Question 3

Consider the region enclosed by the graph of $y = \sqrt{16 - x^2}$ and the x -axis in the domain $0 \leq x \leq 4$.



The area of the shaded quadrant $= \frac{1}{4} \times \pi \times 4^2 = 4\pi$ units².

Therefore, option B.

Question 4

$$T_9 = S_9 - S_8 = 92 - 50 = 42$$

Therefore, option B.

Question 5

Since the annuity is compounded every six months, the rate is 2% per period over 6 periods.

$$\begin{aligned}FV &= a \times X \\&= a \times 6.308 \\&= \$34\,694\end{aligned}$$

Therefore, option C.

Question 6

$$\begin{aligned}g(-4) &= -g(4) \\&= -1\end{aligned}$$

$$\begin{aligned}f(-1) &= f(1) \\&= 3\end{aligned}$$

Therefore, option B.

Question 7

The graph of $y = g(x)$ is obtained using the graph of $y = f(x)$ through a reflection about the x -axis, and a horizontal dilation of factor 2.

Hence, A' has coordinates $(8, -2)$.

Therefore, option D.

Question 8

$$\begin{aligned}\frac{dP}{dt} &= -0.32 \times P_0 e^{-0.32t} \\&= -0.32P \\&= -0.32 \times 4150 \quad \text{since } P = 4150 \text{ when } t = 15 \\ \therefore \frac{dP}{dt} &= -1328\end{aligned}$$

Therefore, option B.

Question 9

Let there be 30 items in the sample size. We can represent the information given in the two-way table as shown:

	R	$\bar{R}$	Total
S	6		18
$\bar{S}$			
Total	12		30

Filling in the empty cells gives:

	R	$\bar{R}$	Total
S	6	12	18
$\bar{S}$	6	6	12
Total	12	18	30

$$\text{Hence } P(R \cup S) = \frac{6 + 6 + 12}{30} = \frac{4}{5}.$$

Therefore option D.

Question 10

The x -coordinates of the maximum turning points forms an arithmetic progression:
 $2, 10, 18, \dots, 50$.

Hence, there are 7 maximum turning points in the domain $0 \leq x \leq 50$.

The graph of $y = \ln x$ is a monotonically increasing function with a maximum value of 3.91 in the domain $0 \leq x \leq 50$, correct to two decimal places.

From the diagram, we can deduce that the graphs of $y = 4 \sin\left(\frac{\pi}{4}x\right)$ and $y = \ln x$ intersect on either side of each maximum turning point on the graph of $y = 4 \sin\left(\frac{\pi}{4}x\right)$ from $x = 10$ to $x = 42$. The graphs meet once right after the maximum turning point at $x = 2$ and again before the turning point at $x = 50$.

Hence, the graphs meet $5 \times 2 + 2 = 12$ times.

Therefore option C.

Section II**Question 11 (a)**

Marking Criteria	Marks
• Provides correct solution	2
• Uses the sine rule, or equivalent merit	1

$$\frac{AC}{\sin 50^\circ} = \frac{5}{\sin 70^\circ}$$

$$AC = \frac{5 \times \sin 50^\circ}{\sin 70^\circ}$$

$$\therefore AC = 4.08 \text{ cm (to 2 d.p.)}$$

Question 11 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the size of $\angle CAB$, or equivalent merit	1

$$\text{Area} = \frac{1}{2} \times 4.08 \times 5 \times \sin 60^\circ$$

$$= 9 \text{ cm}^2 \text{ (to the nearest cm)}$$

Question 12 (a)

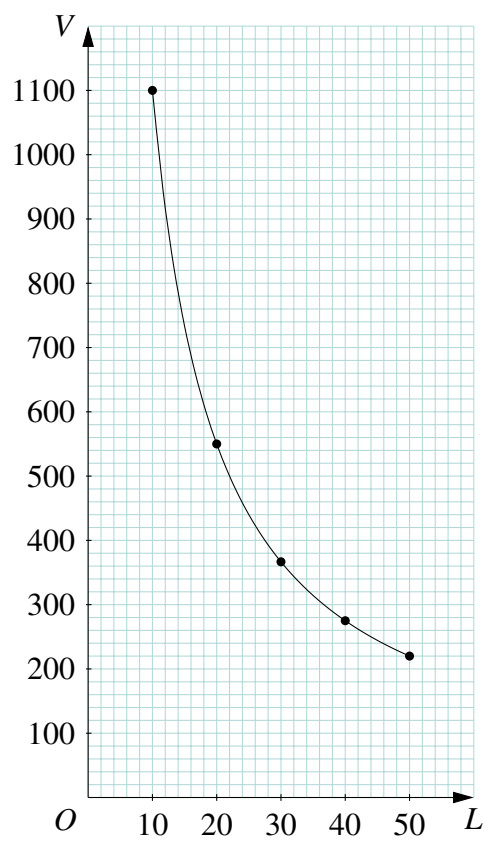
Marking Criteria	Marks
• Provides correct solution	2
• Finds the constant of proportionality, or equivalent merit	1

$$\begin{aligned}V &\propto \frac{1}{L} \\V &= \frac{k}{L} \\550 &= \frac{k}{20} \\k &= 11000 \\\therefore V &= \frac{11000}{L}\end{aligned}$$

Question 12 (b)

Marking Criteria	Marks
• Provides correct solution	2
• At least three values in the table are correct	1

L	10	20	30	40	50
V	1100	550	366.67	275	220



Question 13

Marking Criteria	Marks
• Provides correct solution	2
• Finds the number of terms in the series, or equivalent merit	1

The series is arithmetic with first term $a = 17$ and common difference $d = -3$.

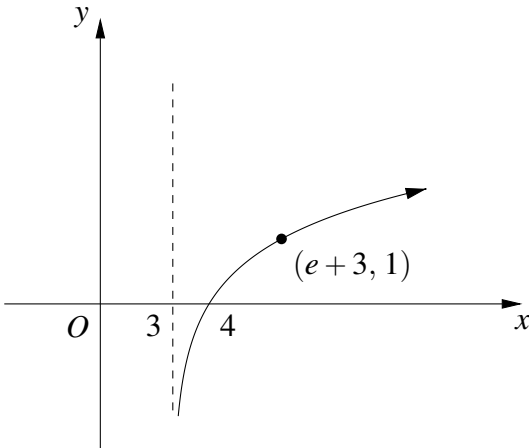
$$-106 = 17 + (n - 1) \times -3$$

$$\therefore n = 42$$

$$\text{Hence } S_{42} = \frac{42}{3} [17 + (-106)] = -1869.$$

Question 14 (a)

Marking Criteria	Marks
Provides correct solution	2
Draws a graph with ONE correct feature. e.g x-intercept, vertical asymptote	1



Question 14 (b)

Marking Criteria	Marks
Provides correct solution	2
Finds the width of each sub-interval, or equivalent merit	1

x	4	4.5	5	5.5	6
y	0	0.405	0.693	0.916	1.098

$$\begin{aligned}
 \text{Area} &\approx \frac{0.5}{2} \left[(0 + 1.098) + 2(0.405 + 0.693 + 0.916) \right] \\
 &= 1.507 \text{ units}^2 \\
 &= 1.5 \text{ units}^2 \quad (\text{to 1 d.p.})
 \end{aligned}$$

Question 15

Marking Criteria	Marks
• Provides correct solution	2
• Makes progress towards solution	1

$$\begin{aligned}
 \int_1^4 \sqrt{3x-2} dx &= \int_1^4 (3x-2)^{\frac{1}{2}} dx \\
 &= \left[\frac{(3x-2)^{\frac{3}{2}}}{\frac{3}{2} \times 3} \right]_1^4 \\
 &= 6.81
 \end{aligned}$$

Question 16

Marking Criteria	Marks
• Provides correct solution	3
• Makes progress towards solution	2
• Applies a valid logarithm law	1

$$\begin{aligned}
 9 \log_5 x &= 25 \log_x 5 \\
 9 \frac{\log x}{\log 5} &= 25 \frac{\log 5}{\log x} \\
 9 (\log x)^2 &= 25 (\log 5)^2 \\
 \log x &= \pm \frac{5}{3} \log 5 \\
 \therefore x &= 5^{\frac{5}{3}}, \quad x = 5^{-\frac{5}{3}}
 \end{aligned}$$

Question 17

Marking Criteria	Marks
• Provides correct solution	3
• Integrates the function and attempts to evaluate their result	2
• Finds a value for one of the integral provided in the question	1

$$\begin{aligned}
 \int_0^{\frac{\pi}{3}} f(x) dx &= \left[-\cos\left(x - \frac{\pi}{3}\right) \right]_0^{\frac{\pi}{3}} \\
 &= -\frac{1}{2} \\
 \therefore \int_{\frac{\pi}{3}}^0 f(x) dx &= \frac{1}{2} \\
 \text{and } \int_{\frac{\pi}{3}}^k f(x) dx &= \frac{3}{2} \\
 \left[-\cos\left(x - \frac{\pi}{3}\right) \right]_{\frac{\pi}{3}}^k &= \frac{3}{2} \\
 \cos\left(x - \frac{\pi}{3}\right) &= -\frac{1}{2} \\
 \therefore k &= \pi
 \end{aligned}$$

Question 18 (a)

Marking Criteria	Marks
• Provides correct solution	4
• Uses a valid method to determine the nature of the stationary points	3
• Finds the x -coordinate of the stationary points, or equivalent merit	2
• Finds the gradient function, or equivalent merit	1

$$\begin{aligned}
 f'(x) &= \frac{1}{3} \left[(x+1) \times 2 \times (x-4) + (x-4)^2 \right] \\
 &= \frac{1}{3} (x-4) \left[2x+2+x-4 \right] \\
 f'(x) &= \frac{1}{3} (x-4) (3x-2)
 \end{aligned}$$

let $f'(x) = 0$,

i.e. $\frac{1}{3} (x-4) (3x-2) = 0$

$\therefore x = 4, \quad x = \frac{2}{3}$

$f(4) = 0, \quad f\left(\frac{2}{3}\right) = \frac{500}{81} \approx 6.173$

$f''(x) = \frac{1}{3} (6x-14)$

$f''(4) = \frac{1}{3} (10) > 0 \Rightarrow (4, 0)$ is a minimum turning point.

$f''\left(\frac{2}{3}\right) = \frac{1}{3} (-10) < 0 \Rightarrow \left(\frac{2}{3}, \frac{500}{81}\right)$ is a maximum turning point.

Question 18 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the coordinates of the possible point of inflexion.	1

Let $f''(x) = 0$

i.e. $\frac{1}{3}(6x - 14) = 0$

$\therefore x = \frac{7}{3} \Rightarrow y = \frac{250}{81}$

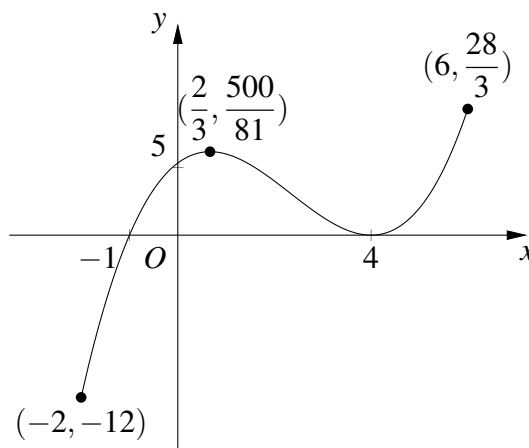
Test:

x	2.3	$\frac{7}{3}$	2.4
$f''(x)$	-0.07	0	0.13
$f'(x)$	-2.8	-2.8	-2.8

Hence, $\left(\frac{7}{3}, \frac{250}{81}\right)$ is a point of inflexion.

Question 18 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Draws a graph showing some of the features from parts (a) or (b)	1



Question 18 (d)

Marking Criteria	Marks
• Provides correct solution	1

$\frac{28}{3}$

Marking Guidelines continue on next page

Question 19 (a)

Marking Criteria	Marks
• Provides correct solution	1

On average, the hourly rate of an employee increases by \$1.189 with each year of experience gained.

Question 19 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the value of c	1

$$33.20 = 1.189(11) + c$$

$$c = 20.12$$

$$\therefore y = 1.189x + 20.12$$

when $x = 10$,

$$y = 1.189(10) + 20.12$$

$$= 32.01$$

An employee with 10 years experience will earn approximately \$32.01 per hour.

Question 19 (c)

Marking Criteria	Marks
• Provides correct solution	1

Since the correlation coefficient of the second dataset (0.850) is smaller than the correlation coefficient (0.921) of the first dataset, it is therefore better to use the first dataset to predict the relationship between hourly wage and years of experience.

Question 20 (a)

Marking Criteria	Marks
• Provides correct solution	1

$$\begin{aligned}
 \text{LHS} &= \frac{\sec^2 x}{\tan^2 x} \\
 &= \frac{1}{\cos^2 x} \div \frac{\sin^2 x}{\cos^2 x} \\
 &= \frac{1}{\cos^2 x} \times \frac{\cos^2 x}{\sin^2 x} \\
 &= \frac{1}{\sin^2 x} \\
 &= \operatorname{cosec}^2 x \\
 &= \text{RHS}
 \end{aligned}$$

Question 20 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Uses the result from part (a)	1

$$\begin{aligned}
 \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \operatorname{cosec}^2 x \, dx &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sec^2 x}{\tan^2 x} \, dx \\
 &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \sec^2 x (\tan x)^{-2} \, dx \\
 &= \left[-\frac{1}{\tan x} \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}} \\
 &= -\frac{1}{\sqrt{3}} + \frac{1}{\frac{1}{\sqrt{3}}} \\
 &= \sqrt{3} - \frac{1}{\sqrt{3}}
 \end{aligned}$$

Question 21 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Uses a valid method to list the sample space	1

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

Therefore $P(X = 3) = \frac{1}{6}$.

Question 21 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Makes progress towards solution	1

$$E(X) = \left(1 \times \frac{10}{36}\right) + \left(2 \times \frac{8}{36}\right) + \left(3 \times \frac{6}{36}\right) + \left(4 \times \frac{4}{36}\right) + \left(5 \times \frac{2}{36}\right)$$

$$\therefore E(X) = \frac{35}{36} \approx 1.94$$

Question 21 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Makes progress towards solution	1

Let σ_X denote the standard deviation of X .

$$\begin{aligned}\text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \frac{35}{6} - \left(\frac{35}{18}\right)^2 \\ \text{Var}(X) &= \frac{665}{324} \\ \therefore \sigma_X &\approx 1.43\end{aligned}$$

Question 22 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the gradient of ell_1	1

$$f'(x) = -\frac{k}{x^2}$$

$$f'(p) = -\frac{k}{p^2}$$

i.e. $y - \frac{y}{p} = -\frac{k}{p^2}(x - p)$

$$p^2y - pk = -kx + kp$$

$$\therefore kx + p^2y = 2pk$$

Question 22 (b)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the x OR y -intercept of ell_1	1

$$x\text{-intercept} = 2p$$

$$y\text{-intercept} = \frac{2k}{p}$$

$$\text{Area} = \frac{1}{2} \times 2p \times \frac{2k}{p} = 2k \text{ units}^2$$

Question 22 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the area of the rectangle	1

$$a = 4 \implies p = 2.$$

Hence, the area of the rectangle is 12 units^2 .

i.e. $2k = 12 \implies k = 6$

Question 23 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the correct value from the table	1

$$\begin{aligned}
 PV &= a \times X \\
 a &= \frac{PV}{X} \\
 &= \frac{350\,000}{155.2069} \\
 &= \$2255.05 \quad (\text{nearest cent})
 \end{aligned}$$

Question 23 (b)

Marking Criteria	Marks
• Provides correct solution	1

$$\begin{aligned}
 A &= 350\,000 \times \frac{6\%}{12} \\
 &= \$1750
 \end{aligned}$$

Question 23 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the total repaid, or equivalent merit	1

Interest = total repayments – amount borrowed

$$\begin{aligned}
 I &= 2255.05 \times (25 \times 12) - 350\,000 \\
 &= 676\,515 - 350\,000 \\
 &= \$326\,515
 \end{aligned}$$

Noah pays \$326 515 in interest.

Question 23 (d)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the total repaid with the second scenario, or equivalent merit	1

15 years = 180 months.

$$\begin{aligned}
 \text{total regular repayments} &= (180 + 86) \times 2255.05 \\
 &= \$599\,843.30 \\
 \therefore \text{total repaid} &= \$599\,843.30 + 50\,000 \\
 &= \$649\,843.30 \\
 \text{interest saved} &= \$676\,515 - \$649\,843.30 \\
 &= \$26\,671.70
 \end{aligned}$$

Noah saves \$26 671.70 interest by making the lump sum payment of \$50 000.

Question 24 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Finds either c OR k	1

$c = 10$ and $k = 4$.

Question 24 (b)

Marking Criteria	Marks
• Provides correct solution	1

$$\frac{2\pi}{b} = 24$$

$$\therefore b = \frac{\pi}{12}$$

Question 24 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Finds a time when the water depth is 11 metres	1

$$10 + 4 \sin\left(\frac{\pi}{12}t\right) = 11$$

$$\sin\left(\frac{\pi}{12}t\right) = \frac{1}{4}$$

$$\frac{\pi}{12}t = \sin^{-1}\left(\frac{1}{4}\right), \pi - \sin^{-1}\left(\frac{1}{4}\right)$$

$$\therefore t = 0.965, 11.035$$

The depth of water in the reservoir is 11 metres or more for $11.035 - 0.965 = 10.1$ hours.

Question 25 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Finds $P(X \leq 3)$ or $P(X \leq 1)$	1

$$\begin{aligned}
 P(1 \leq X \leq 3) &= P(X \leq 3) - P(X \leq 1) \\
 &= \frac{1}{16} \times 3^2 \times (5 - 2\sqrt{3}) - \frac{1}{16} \times 1^2 \times (5 - 2\sqrt{1}) \\
 &= 0.86394 \dots - \frac{3}{16} \\
 &= 0.68 \quad (2 \text{ d.p.})
 \end{aligned}$$

Question 25 (b)

Marking Criteria	Marks
• Provides correct solution	3
• Makes progress towards solution	2
• Finds an expression for the probability density function	1

$$\begin{aligned}
 F(x) &= \frac{1}{16}x^2(5 - 2\sqrt{x}) \\
 &= \frac{1}{16}\left(5x^2 - 2x^{\frac{5}{2}}\right)
 \end{aligned}$$

$$\begin{aligned}
 f(x) &= \frac{1}{16}\left(10x - 5x^{\frac{3}{2}}\right) \\
 f'(x) &= \frac{1}{16}\left(10 - \frac{15}{2}x^{\frac{1}{2}}\right)
 \end{aligned}$$

Let $f'(x) = 0$

$$\begin{aligned}
 \frac{1}{16}\left(10 - \frac{15}{2}x^{\frac{1}{2}}\right) &= 0 \\
 x^{\frac{1}{2}} &= \frac{20}{15} \\
 \therefore x &= \frac{16}{9}
 \end{aligned}$$

Question 26

Marking Criteria	Marks
• Provides correct solution	4
• Makes significant progress towards solution	3
• Finds an expression for the shaded area	2
• Finds the x -coordinate point of intersection	1

$$e^{-x} + 2 = -e^{-x} + 3$$

$$2e^{-x} = 1$$

$$e^{-x} = \frac{1}{2}$$

$$\therefore x = \ln 2$$

$$\text{Area} = \int_0^{\ln 2} (e^{-x} + 2) - (-e^{-x} + 3) \, dx$$

$$= \int_0^{\ln 2} 2e^{-x} - 1 \, dx$$

$$= \left[-2e^{-x} - x \right]_0^{\ln 2}$$

$$= (-2e^{-\ln 2} - \ln 2) - (-2e^0 - 0)$$

$$\therefore \text{Area} = 1 - \ln 2 \text{ units}^2.$$

Question 27

Marking Criteria	Marks
• Provides correct solution	4
• Makes significant progress towards solution	3
• Establishes a recurrence relationship involving a geometric series	2
• Finds the amount of herbs in the garden after one week, or equivalent merit	1

Let W_n be the volume, in kg, of herbs in the garden after the n^{th} week.

$$\begin{aligned}
 W_1 &= 10(1.15) - 0.5 \\
 W_2 &= 10(1.15)^2 - 0.5(1.15) - 0.5 \\
 W_3 &= 10(1.15)^3 - 0.5(1.15)^2 - 0.5(1.15) - 0.5 \\
 &= 10(1.15)^3 - 0.5(1 + 1.15 + 1.15^2) \\
 &\vdots \\
 W_n &= 10(1.15)^n - 0.5(1 + 1.15 + 1.15^2 + \dots + 1.15^{n-1}) \\
 \therefore W_n &= 10(1.15)^n - 0.5 \times \frac{1.15^n - 1}{0.15}
 \end{aligned}$$

The farmer needs to fulfil a 50 kg order after n weeks, so let $W_n = 50$.

$$\begin{aligned}
 10(1.15)^n - 0.5 \times \frac{1.15^n - 1}{0.15} &= 50 \\
 10(1.15)^n - \frac{10}{3}(1.15^n - 1) &= 50 \\
 \frac{20}{3}(1.15)^n &= \frac{140}{3} \\
 1.15^n &= 7 \\
 n &= \frac{\log 7}{\log 1.15} = 13.92\dots \\
 \therefore \text{The order can be fulfilled after 14 weeks.}
 \end{aligned}$$

Question 28

Marking Criteria	Marks
• Provides correct solution	3
• Makes progress towards solution	2
• Finds the corresponding z -score	1

$$P(Z \leq z) = 0.67 \Rightarrow z = 0.44$$

$$\begin{aligned}\text{i.e. } 0.44 &= \frac{X - 35}{5} \\ X &= 0.44 \times 5 + 35 \\ \therefore X &= 37.2\end{aligned}$$

Question 29 (a)

Marking Criteria	Marks
• Provides correct solution	1

$$\begin{aligned}\frac{x^2}{4} + \frac{y^2}{9} &= 1 \\ \frac{y^2}{9} &= 1 - \frac{x^2}{4} \\ y^2 &= 9 - \frac{9}{4}x^2 \\ \therefore y &= \sqrt{9 - \frac{9}{4}x^2} \quad \text{since } y > 0.\end{aligned}$$

Question 29 (b)

Marking Criteria	Marks
• Provides correct solution	4
• Makes significant progress towards solution	3
• Finds the expression for the rate of change in the shaded area with respect to the variable	2
• Finds an expression for the shaded area in terms of one variable	1

Let B have coordinates (x, y) and R denote the area of the shaded region.

$$\begin{aligned}
 R &= \frac{1}{2} \times (2+x) \times 2y \\
 &= y(2+x) \\
 \therefore R &= (2+x) \sqrt{9 - \frac{9}{4}x^2}
 \end{aligned}$$

Let $\frac{dR}{dx} = 0$

$$\begin{aligned}
 \text{i.e. } \frac{dR}{dx} &= \sqrt{9 - \frac{9}{4}x^2} - \frac{9}{4}x(2+x) \left(9 - \frac{9}{4}x^2\right)^{-\frac{1}{2}} \\
 0 &= \sqrt{9 - \frac{9}{4}x^2} - \frac{9x(2+x)}{4\sqrt{9 - \frac{9}{4}x^2}} \\
 \frac{9x(2+x)}{4\sqrt{9 - \frac{9}{4}x^2}} &= \sqrt{9 - \frac{9}{4}x^2} \\
 9x(2+x) &= 4 \left(9 - \frac{9}{4}x^2\right) \\
 18x + 9x^2 &= 36 - 9x^2 \\
 x^2 + x - 2 &= 0 \\
 \therefore x &= 1 \quad (x \neq -2 \text{ since there would be no triangle in this case.})
 \end{aligned}$$

Testing the solution:

x	0.9	1	1.1
$\frac{dR}{dx}$	0.49	0	-0.56

Hence, the shaded area will be a maximum when $x = 1$.

Question 30 (a)

Marking Criteria	Marks
• Provides correct solution	2
• Finds the anti-derivative of $\ddot{x}$, or equivalent merit	1

$$\dot{x} = -60 \cos(2t) + c$$

$$\therefore \dot{x} = -60 \cos(2t) + 130 \quad \text{since } \dot{x} = 70 \text{ when } t = 0$$

Question 30 (b)

Marking Criteria	Marks
• Provides correct solution	3
• Makes progress towards finding the distance travelled	2
• Finds when the velocity of the particle first reaches 160 km/h, or equivalent merit	1

$$-60 \cos(2t) + 130 = 160$$

$$\cos(2t) = -\frac{1}{2}$$

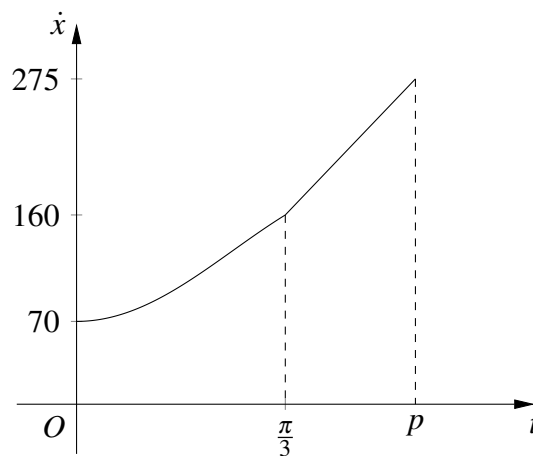
$$2t = \frac{2\pi}{3}$$

$$\therefore t = \frac{\pi}{3} \text{ seconds}$$

$$\begin{aligned} \text{Distance travelled} &= \int_0^{\frac{\pi}{3}} -60 \cos(2t) + 130 \, dt \\ &= \left[-30 \sin \frac{2\pi}{3} + 130t \right]_0^{\frac{\pi}{3}} \\ &= -30 \sin \frac{2\pi}{3} + \frac{120\pi}{3} \\ &= 110.15 \text{ km (to 2 d.p.)} \end{aligned}$$

Question 30 (c)

Marking Criteria	Marks
• Provides correct solution	2
• Recognises that the region enclosed by the velocity-time graph in the second stage is trapezoidal	1



Let p be the time when the particle reaches the required velocity 275 km/h.

The area under the graph in the second stage equals to the area under the graph under the first stage.

$$110.15 = \frac{p - \frac{\pi}{3}}{3} (160 + 275)$$

$$\therefore p = 1.8$$